

ECO 329 ECONOMIC STATISTICS (36110)

Lecture Notes

Fall 2025

4 Random Variables

4.1 Basics

Definition 4.1.1.0. A **random variable** (r.v.) is a function that maps elements of the sample space into the set of real numbers.

- A random variable characterizes an experiment by associating a real number - probability - with each outcome in the sample space.
- For a random variable X with sample space $\mathbb{S}$, $X(\omega) = p \in \mathbb{R}$ for some $\omega \in \mathbb{S}$ means that, in an experiment characterized by X , the probability that ω occurs is p .

Definition 4.1.1.1. A random variable is **discrete** if its associated sample space is countable.

Definition 4.1.1.2. A random variable is **continuous** if its associated sample space consists of only intervals.

- The discussion of continuous r.v.'s usually focus on sample spaces constituted by real numbers, which justifies the definition above that involves intervals.
- When a sample space is constituted by real numbers, it is uncountable iff it consists of an interval. However, a r.v. need not be continuous if its sample space consists of intervals - it may be "mixed".¹

4.2 Discrete Random Variables

In sections 4.2.1 and 4.2.2, unless specified otherwise, let X, Y be two discrete r.v.'s defined on sample spaces $\mathbb{S}_X, \mathbb{S}_Y$ respectively.

¹Mixed r.v. is not required in this course.

4.2.1 Characterization

Definition 4.2.1.1. The probability distribution or **probability mass function** (pmf) of X , denoted as $P_X(\cdot)$, is a function such that

$$P_X(x) = P(X = x) \text{ for all } x \in \mathbb{S}_X.$$

- In other words, pmf gives the probability of every outcome in the sample space.
- A pmf $P_X(\cdot)$ is well-defined iff it satisfies the following two conditions:
 - $P_X(x) \geq 0$ for all $x \in \mathbb{S}_X$.
 - $\sum_{x \in \mathbb{S}_X} P_X(x) = 1$.

Definition 4.2.1.2. The **joint probability mass function** of X and Y - or (X, Y) - denoted as $P_{XY}(\cdot, \cdot)$, is a function such that

$$P_{XY}(x, y) = P(X = x \text{ and } Y = y) \text{ for all } x \in \mathbb{S}_X \text{ and all } y \in \mathbb{S}_Y.$$

- Similar to the univariate situation, $P_{XY}(\cdot, \cdot)$ is well-defined iff
 - $P_{XY}(x, y) \geq 0$ for all $x \in \mathbb{S}_X$ and all $y \in \mathbb{S}_Y$, and
 - $\sum_{x \in \mathbb{S}_X} \sum_{y \in \mathbb{S}_Y} P_{XY}(x, y) = 1$.

Definition 4.2.1.3. The **marginal probability mass function** of X , denoted as $P_X(\cdot)$, is given by

$$P_X(x) = \sum_{y \in \mathbb{S}_Y} P_{XY}(x, y).$$

Definition 4.2.1.4. Suppose a pmf $P_X(\cdot)$ depends on a quantity that can be assigned to be any element in a set of possible values, with each different value resulting in a different probability distribution. Such a quantity is called a **parameter** of the distribution. The collection of all such probability distributions with different values of the parameter is called a **family of probability distributions**.

- Back to our simplest example of coin toss and assume that the coin is not necessarily fair. Then, the pmf of the coin toss can be written as $P(H; p) = p$ and $P(T; p) = 1 - p$, where $p \in [0, 1]$ is a parameter and $P(\cdot; p)$ characterizes a family of probability distributions.

Definition 4.2.1.5. X and Y are said to be **independent** if

$$P_{XY}(x, y) = P_X(x)P_Y(y) \text{ for all } x \in \mathbb{S}_X \text{ and all } y \in \mathbb{S}_Y.$$

- Otherwise, i.e., if $P_{XY}(x, y) \neq P_X(x)P_Y(y)$ for some $x \in \mathbb{S}_X$ and some $y \in \mathbb{S}_Y$, then X and Y are said to be dependent - “dependent” and “independent” are mutually exclusive and exhaustive.

Definition 4.2.1.6. For any $x \in \mathbb{S}_X$ so that $P_X(x) > 0$, the **conditional probability mass function** of Y given $X = x$, denoted as $P_{Y|X}(\cdot|x)$, is given by

$$P_{Y|X}(y|x) = \frac{P_{XY}(x, y)}{P_X(x)} \text{ for all } y \in \mathbb{S}_Y.$$

Definition 4.2.1.7. Let $\mathbb{S}_X \subset \mathbb{R}$. The **cumulative distribution function** (CDF) of X , denoted as $F_X(\cdot)$, is given by

$$F_X(x) = P(X \leq x) = \sum_{z \leq x} P_X(z) \text{ for all } x \in \mathbb{S}_X.$$

- That is, for every number $x \in \mathbb{R}$, CDF gives the probability that the observed value of a random variable is no more than the number x .
- A CDF $F_X(\cdot)$ is well-defined iff it satisfies the following three conditions:
 - $F_X(x) \leq F_X(x')$ for all $x, x' \in \mathbb{R}$ so that $x < x'$.
 - $\lim_{x \rightarrow -\infty} F_X(x) = 0$ and $\lim_{x \rightarrow +\infty} F_X(x) = 1$.
 - For any $x_0 \in \mathbb{R}$, $\lim_{x \downarrow x_0} F_X(x) = F_X(x_0)$ - that is, $F_X(\cdot)$ is right-continuous.

Definition 4.2.1.8. Let $\mathbb{S}_X, \mathbb{S}_Y \subset \mathbb{R}$. The **joint cumulative distribution function** of (X, Y) , denoted as $F_{XY}(\cdot, \cdot)$, is given by

$$F_{XY}(x, y) = P(X \leq x \text{ and } Y \leq y) = \sum_{z \leq x} \sum_{u \leq y} P_{XY}(z, u) \text{ for all } x \in \mathbb{S}_X \text{ and all } y \in \mathbb{S}_Y.$$

Proposition 4.2.1.9. Let $\mathbb{S}_X \subset \mathbb{R}$, then for any $a, b \in \mathbb{R}$ with $a \leq b$,

1. $P\{x \in [a, b]\} = F_X(b) - \lim_{x \uparrow a} F_X(x)$,
2. $P\{x \in (a, b]\} = F_X(b) - F_X(a)$,
3. $P\{x \in [a, b)\} = \lim_{x \uparrow b} F_X(x) - \lim_{x \uparrow a} F_X(x)$, and
4. $P\{x \in (a, b)\} = \lim_{x \uparrow b} F_X(x) - F_X(a)$.

- All univariate/bivariate definitions and results above can be extended to multivariate situations - in ways similar to what we have done to extend univariate things to bivariate situations.

4.2.2 Statistics

Throughout this section, let $\mathbb{S}_X, \mathbb{S}_Y \subset \mathbb{R}$.

Definition 4.2.2.1.1. The **expected value**, or mean value, of X , denoted as $E(X)$ - or μ_X (or just μ in univariate context) - is

$$E(X) = \sum_{x \in \mathbb{S}_X} x P_X(x).$$

- The expected value of a random variable $h(X)$, which is a function of X , is

$$E[h(X)] = \sum_{x \in \mathbb{S}_X} h(x) P_X(x).$$

- Similarly, the expected value of a random variable $h(X, Y)$, which is a function of X and Y , is

$$E[h(X, Y)] = \sum_{x \in \mathbb{S}_X} \sum_{y \in \mathbb{S}_Y} h(x, y) P_{XY}(x, y).$$

Definition 4.2.2.1.2. For a sample $\{x_i\}_{i=1}^n$, the **sample mean** of the variable X , denoted by $\bar{x}$, is

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}.$$

Definition 4.2.2.2.1. The q^{th} **percentile** of X is the value of x so that

$$F_X(x) \geq q\% \text{ and } P(X \geq x) \geq q\%.$$

- $P(X \geq x) \geq q\%$ can be equivalently written as

$$\lim_{x' \uparrow x} F_X(x') \leq q\%.$$

- When $q = 50$, the above value of x is also called the **median** of X and usually denoted as $m(X)$.
- When $q = 25$ (75), the above value of x is also called the **first (third) quartile** of X .

Definition 4.2.2.2.2. The q^{th} **percentile** of a sample $\{x_i\}_{i=1}^n$ is an observation $x \in \{x_i\}_{i=1}^n$ so that the fraction of observations in the sample that is no larger than x is $q\%$.

- In practice, when the sample $\{x_i\}_{i=1}^n$ is ordered so that $x_j \leq x_{j+1}$ for each $j \in \{1, \dots, n-1\}$, then the q^{th} percentile of $\{x_i\}_{i=1}^n$ is

$$x_{\frac{q(n+1)}{100}}.$$

- Or, in situations where $\frac{q(n+1)}{100}$ is not an integer, the percentile is a weighted average of x_k and x_{k+1} with $k \in \mathbb{N}_+$ and $k < \frac{q(n+1)}{100} < k+1$.

Definition 4.2.2.3.1. The **variance** of X , denoted as $V(X)$ - or σ_X^2 (or just σ^2 in univariate context) - is

$$V(X) = \sum_{x \in \mathbb{S}_X} [x - E(X)]^2 P_X(x) = E\{[X - E(X)]^2\},$$

and the **standard deviation** of X is $\sigma_X = \sqrt{V(X)}$.

- The variance of a random variable $h(X)$ is

$$V[h(X)] = \sum_{x \in \mathbb{S}_X} \{h(x) - E[h(x)]\}^2 P_X(x).$$

Definition 4.2.2.3.2. For a sample $\{x_i\}_{i=1}^n$, the **sample variance** of the variable X , denoted by s_X^2 (or just s^2 in univariate context), is

$$s_X^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1},$$

and $s = \sqrt{s^2}$ is the **sample standard deviation**.

Proposition 4.2.2.4.

$$V(X) = E(X^2) - [E(X)]^2.$$

- Proof idea: Direct calculation by definitions.
- Also from direct calculation, the counterpart of the above alternative formula doesn't directly work for sample variance - though it could be used for calculating $\sum_{i=1}^n (x_i - \bar{x})^2$.

Definition 4.2.2.5. For any $a, b \in \mathbb{R}$, the new random variable $aX + b$ is called a **linear transformation** of X .

Proposition 4.2.2.6. For any $a, b \in \mathbb{R}$,

$$E(aX + b) = aE(X) + b, \quad m(aX + b) = am(X) + b, \quad \text{and} \quad V(aX + b) = a^2V(X).$$

- Proof idea: Direct calculation by definitions.
- Proposition 4.2.2.6 can be extended to general percentiles and its sample-wise counterpart.

Definition 4.2.2.7.1. The **covariance** between X and Y , denoted as $Cov(X, Y)$, is

$$Cov(X, Y) = E\{[X - E(X)][Y - E(Y)]\} = \sum_{x \in \mathbb{S}_X} \sum_{y \in \mathbb{S}_Y} [x - E(X)][y - E(Y)]P_{XY}(x, y).$$

Definition 4.2.2.7.2. For a bivariate sample $\{(x_i, y_i)\}_{i=1}^n$, the **sample covariance** between the two variables X and Y , denoted as s_{XY} , is

$$s_{XY} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}.$$

- Note that the variance of a variable is the covariance between it and itself.

Proposition 4.2.2.8.

$$Cov(X, Y) = E(XY) - E(X)E(Y).$$

- Proof idea: Direct calculation by definitions.
- Similar to the case for variance, the counterpart of the above alternative formula doesn't directly work for sample covariance.

Definition 4.2.2.9.1. The **correlation** between X and Y , denoted as $Corr(X, Y)$ - or ρ_{XY} (or just ρ in a bivariate context) - is

$$Corr(X, Y) = \frac{Cov(X, Y)}{\sqrt{V(X)V(Y)}}.$$

Definition 4.2.2.9.2. For a bivariate sample $\{(x_i, y_i)\}_{i=1}^n$, the **sample correlation** between the two variables X and Y , denoted as r_{XY} , is

$$r_{XY} = \frac{s_{XY}}{s_X s_Y}.$$

- The value of (sample) correlation is always between -1 and 1.
 - Key fact used to prove this result: $0 \leq V\left(\frac{X}{\sigma_X} \pm \frac{Y}{\sigma_Y}\right)$.
- Both covariance and correlation only measure the degree of linear relationship between two (random) variables.

Proposition 4.2.2.10. $|\rho_{XY}| = 1$ iff $Y = aX + b$ for some $a, b \in \mathbb{R}$ with $a \neq 0$.

- Proof idea:
 - If direction: Direct calculation by definitions.
 - Only if direction: Again make use of $V\left(\frac{X}{\sigma_X} \pm \frac{Y}{\sigma_Y}\right)$ and the fact that a r.v. W is with 0 variance iff $P(W = c) = 1$ for some constant $c \in \mathbb{R}$.
 - * In the context of this course - or if we only consider discrete r.v.'s - $P(W = c) = 1$ is equivalent to that the r.v. W degenerates into the deterministic constant c .

Definition 4.2.2.11. X and Y are said to be **uncorrelated** if $\rho_{XY} = 0$.

Proposition 4.2.2.12. If X and Y are independent, then $\rho_{XY} = 0$.

- Proof idea: Direct calculation by definitions, aiming at showing $Cov(X, Y) = 0$.
- By contrast, zero correlation doesn't imply independence - intuitively, $\rho_{XY} = 0$ actually means zero linear correlation while doesn't rule out possible nonlinear relationships.

Proposition 4.2.2.13. For any $a, b, c, d \in \mathbb{R}$,

1. $Cov(aX + b, cY + d) = acCov(X, Y)$, and
2. $Corr(aX + b, cY + d) = \begin{cases} \frac{ac}{|ac|} Corr(X, Y) & \text{if } ac \neq 0 \\ 0 & \text{elsewhere} \end{cases}.$

Definition 4.2.2.14. For any given set of random variables $\{X_i\}_{i=1}^n$ and any two given sets of constants $\{a_i\}_{i=1}^n$, $\{b_i\}_{i=1}^n$, the new random variable $\sum_{i=1}^n (a_i X_i + b_i)$ is called a **linear combination** of X_i 's.

Proposition 4.2.2.15. For any two given sets of random variables $\{X_i\}_{i=1}^n$, $\{Y_j\}_{j=1}^n$ and any two given sets of constants $\{a_i\}_{i=1}^n$, $\{b_j\}_{j=1}^n$,

1. $E(\sum_{i=1}^n a_i X_i) = \sum_{i=1}^n a_i E(X_i)$, and
 2. $Cov(\sum_{i=1}^n a_i X_i, \sum_{j=1}^n b_j Y_j) = \sum_{i=1}^n \sum_{j=1}^n a_i b_j Cov(X_i, Y_j)$.
- For both Proposition 4.2.2.13 and Proposition 4.2.2.15, they can be
 - proved by direct calculation with the relevant definitions, and
 - extended to its sample-wise counterpart.

4.2.3 Bernoulli Distribution

Definition 4.2.3.1. A **Bernoulli trial** is an experiment that has two outcomes.

- The two outcomes are usually named as “success” and “failure” respectively.

Definition 4.2.3.2. A **Bernoulli random variable** is a random variable whose only possible values are 0 and 1.

- Put Bernoulli random variable into the context of Bernoulli trial, 1 is usually considered as representing success and 0 is usually considered as representing failure.
- For a Bernoulli random variable X ,
 - its pmf is

$$P_X(x) = \begin{cases} 1-p & \text{if } x = 0 \\ p & \text{if } x = 1 \\ 0 & \text{elsewhere} \end{cases},$$

where $p \in [0, 1]$ is called the **success probability**,

- its CDF is

$$F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1-p & \text{if } x \in [0, 1) \\ 1 & \text{if } x \geq 1 \end{cases},$$

- $E(X) = p$, and $V(X) = p(1-p)$ - both of which are by direct calculation by definitions.

4.2.4 Binomial Distribution

Definition 4.2.4.1. A **binomial experiment** is a sequence of n mutually independent and identical Bernoulli trials with success probability $p \in [0, 1]$, where $n \in \mathbb{N}_+$ is fixed beforehand.

Definition 4.2.4.2. A **binomial random variable** $X \sim b(n, p)$ is a random variable that characterizes the distribution of the number of successes in a binomial experiment consisting of n Bernoulli trials with success probability p .

- For $X \sim b(n, p)$,
 - its pmf is

$$b(x; n, p) = P_X(x) = \begin{cases} C_n^x p^x (1-p)^{n-x} & \text{if } x \in \mathbb{N} \text{ and } x \leq n \\ 0 & \text{elsewhere} \end{cases},$$

- * By the binomial theorem,

$$\sum_{x=0}^n P_X(x) = 1.$$

- $E(X) = np$, and $V(X) = np(1-p)$ - both of which are by Propositions 4.2.2.11 and 4.2.2.13 as well as properties of Bernoulli random variable.

4.2.5 Hypergeometric Distribution

Definition 4.2.5.1. A **hypergeometric random variable** $X \sim H(N, K, n)$ is a random variable that characterizes the distribution of the number of successes in an experiment where n random draws are made from a population of size N consisting of K successes with $n, N, K \in \mathbb{N}$.

- For $X \sim H(N, K, n)$,
 - its pmf is

$$P_X(x) = \begin{cases} \frac{C_K^x C_{N-K}^{n-x}}{C_N^n} & \text{if } x \in \mathbb{N} \text{ and } x \in [\max\{0, n - N + K\}, \min\{n, K\}] \\ 0 & \text{elsewhere} \end{cases},$$

- $E(X) = \frac{nK}{N}$, and $V(X) = \frac{N-n}{N-1} \frac{nK}{N} \left(1 - \frac{K}{N}\right)$.

- * Key fact used in the calculation of $E(X)$ and $V(X)$: $C_a^b = \frac{b}{a} C_{a-1}^{b-1}$.

- Relationships between binomial distribution and hypergeometric distribution:
 - Consider the experiment described in Definition 4.2.5.1 - let's, informally, call it "hypergeometric experiment" - where draws are made at once, i.e., "without replacement". If, instead, the draws are made with replacement so that each of the random draw is made from the population of size N consisting of K successes, the n draws are equivalent to n independent and identical Bernoulli trials with success probability $\frac{K}{N}$. As a result, such n draws with replacement constitute a binomial experiment.

- For a hypergeometric experiment characterized by (N, K, n) , if we consider the n draws to be made sequentially without replacements, the first draw is actually equivalent to a Bernoulli trial with success probability $\frac{K}{N}$, and, for any $m \in \mathbb{N}_+$, the $m+1^{th}$ draw after $s \leq m$ successes is equivalent to a Bernoulli trial with success probability $\frac{K-s}{N-m}$ - which converges to $\frac{K}{N}$ as long as N and K proportionally goes to infinity with $\frac{K}{N}$ unchanged so that s and m are sufficiently small to be neglected. Thus, when the population size N is sufficiently large (say, to the extreme, close to infinity), the hypergeometric experiment characterized by (N, K, n) can actually approximate and be approximated by a binomial experiment characterized by $b(n, \frac{K}{N})$.
- Correspondingly, when we compare the expectations and variances of $H(N, K, n)$ and $b(n, \frac{K}{N})$, the expectations are found to be the same and the only difference in variances is the $\frac{N-n}{N-1}$ term in the variance of hypergeometric random variable, which is usually called **finite population correction factor**, increasing in N , and converges to 1 as $N \rightarrow +\infty$.

4.2.6 Poisson Distribution

Definition 4.2.6.1. A **Poisson random variable** $X \sim Pois(\lambda)$ is a random variable that characterizes the distribution of the number of times an event of interest occurs in an experiment during an interval (continuum) of time.

- In terms of the experiment mentioned in Definition 4.2.6.1. - let's (informally) call it "Poisson experiment", it is assumed to satisfy the following properties:
 1. The number of times the event occurs in any interval is independent of the number of occurrences in any other disjoint interval.
 2. The average number of occurrences per unit time is constant.
 3. The probability of an occurrence in any one interval is proportional to the length of the interval.
 4. The probability of two or more simultaneous occurrences is zero.
- Though the setting associated with Poisson distribution is similar to that regarding binomial distribution, the two settings significantly differ in nature because the latter is about possible number of successes in discrete trials, while the former is about possible number of occurrences in continua of time.
 - In the Poisson setting, there is a trial happening at each time point in a continuum of time, so there are uncountably many trials in such a continuum.
- For $X \sim Pois(\lambda)$,
 - its pmf is

$$p(x; \lambda) = P_X(x) = \begin{cases} \frac{e^{-\lambda} \lambda^x}{x!} & \text{if } x \in \mathbb{N} \\ 0 & \text{elsewhere} \end{cases},$$

- * $\sum_{x \in \mathbb{N}} P_X(x) = 1$ by the Taylor series of $f(x) = e^x$ at $x = 0$.
- $E(X) = V(X) = \lambda$ - both of which are from direct calculation by definitions.
 - * Correspondingly, $\lambda > 0$ is the average number of occurrences (during the interval of time, i.e., per unit time).
- The probability distribution is positively skewed, becoming more symmetric as λ increases.
- The Poisson distribution can be further interpreted by letting $\lambda = \alpha t$, so that it characterizes the occurrences in an experiment during a time interval of length t , where the average number of occurrences per unit time is α .

Proposition 4.2.6.2. As $n \rightarrow +\infty$ and $p \rightarrow 0$ so that $np \rightarrow \lambda > 0$,

$$b(x; n, p) \rightarrow p(x; \lambda) \text{ for all } x \in \mathbb{N}.$$

- Proof: not required.
 - For your reference, a key fact used in the proof is $e = \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}}$, and one proof idea that doesn't require a too strong math background makes use of (1) probability generating function $G(z) = \sum_{x \in \mathbb{N}} P_X(x) z^x$ of a discrete random variable with pmf $P_X(x)$ and (2) binomial theorem.
- Recall that the expectation and variance of $b(n, p)$ are respectively np and $np(1 - p)$, both of which also converges to λ given the conditions stated in Proposition 4.2.6.2.